Sections 3.4, 5.1: Quadratic Spline; Numerical Differentiation.

Problem 2 (20pts) Quadratic Spline

Find a **quadratic spline** interpolation for the data points $\{(x_k, y_k)\} = \{(0,0),(1,2),(2,3),(3,1)\}$.

As an extra condition, require extra smoothness at node x_2: $s_1''(x_2) = s_2''(x_2)$.

Reminder: the quadratic spline will have the form
$$\begin{cases} s_1(x) = y_1 + a_1(x-x_1)^2 + b_1(x-x_1), & x \in [x_1, x_2] \\ s_2(x) = y_2 + a_2(x-x_2)^2 + b_2(x-x_2), & x \in [x_2, x_3] \\ s_3(x) = y_3 + a_3(x-x_3)^2 + b_3(x-x_3), & x \in [x_3, x_4] \end{cases}$$

Since there are **6** unknowns, we need **6** constraints:

- Continuity of the spline (**3** constraints), i.e. correct values at right endpoints: $\boldsymbol{s_k(x_{k+1}) = y_{k+1}}$
- Continuity of spline derivative at all internal nodes (**2** constraints): $\boldsymbol{s_k'(x_{k+1}) = s_{k+1}'(x_{k+1})}$
- **One** extra constraint is needed: impose continuity of 2nd derivative at the 1st internal node: $\boldsymbol{s_1''(x_2) = s_2''(x_2)}$.

Finally, **plot** on separate graphs the spline, its derivative, and its 2nd derivative (by hand is fine).

Hint: first, consider equations for the coefficients of first 2 parts of the spline, $s_1(x)$ and $s_2(x)$. This will immediately reduce to 3 equations for 3 unknowns. After solving this linear system, find the coefficients of the final 3rd piece, $s_3(x)$.

Start by plugging in the given spline points:
$$\begin{cases} s_1(x) = 0 + a_1 x^2 + b_1 x, & x \in [0,1] \\ s_2(x) = 2 + a_2(x-1)^2 + b_2(x-1), & x \in [1,2] \\ s_3(x) = 3 + a_3(x-2)^2 + b_3(x-2), & x \in [2,3] \end{cases}$$

Spline pieces #1 & #2:

The extra condition (see above) $s_1''(x_2) = s_2''(x_2)$ implies that $\boldsymbol{a_1 = a_2}$. Therefore, we only need to find three constants, by matching the values at right endpoints, and enforcing continuity of derivative at x_2:

$$\begin{cases} s_1(1) = a_1 + b_1 = 2, \\ s_2(2) = 2 + \underbrace{a_2}_{=a_1} + b_2 = 2 + a_1 + b_2 = 3, \\ s_1'(1) = s_2'(1): \quad 2a_1 + b_1 = b_2. \end{cases} \Rightarrow \begin{cases} a_1 + b_1 = 2, \\ a_1 + b_2 = 1, \\ 2a_1 + b_1 - b_2 = 0 \end{cases} \Rightarrow \begin{cases} a_1 = a_2 = -\frac{1}{2}, \\ b_1 = 2 - a_1 = \frac{5}{2}, \\ b_2 = 1 - a_1 = \frac{3}{2}. \end{cases}$$

Thus, our first two spline pieces are now complete:
$$\begin{cases} s_1(x) = \frac{-x^2 + 5x}{2}, & x \in [0,1] \\ s_2(x) = \frac{4 - (x-1)^2 + 3(x-1)}{2}, & x \in [1,2] \\ s_3(x) = 3 + a_3(x-2)^2 + b_3(x-2), & x \in [2,3] \end{cases}$$

Spline piece #3:

Next, let's determine the coefficients in the last piece of the spline, $s_3(x)$. Enforcing continuity of 1st derivative across the final internal node $x_3 = 2$, $s_3'(2) = s_2'(2)$, we obtain:

$$b_3 = 2a_2 + b_2 = -1 + \frac{3}{2} = \frac{1}{2}$$

Finally, let's fit the value of $y_3 = 1$ at right end-point $x_4 = 3$:

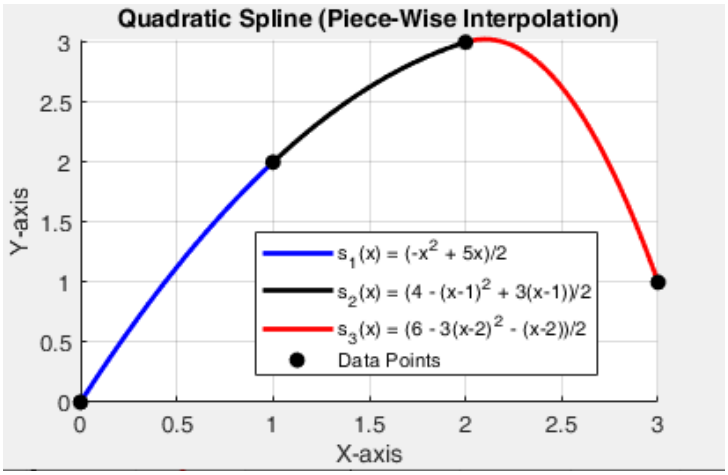
$$s_3(3) = 3 + a_3 + b_3 = \frac{7}{2} + a_3 = y_4 = 1$$

From this we obtain $\boldsymbol{a_3 = -\frac{5}{2}}$. Thus, we find that

$$s_3(x) = \frac{6 - 5(x-2)^2 + (x-2)}{2}$$

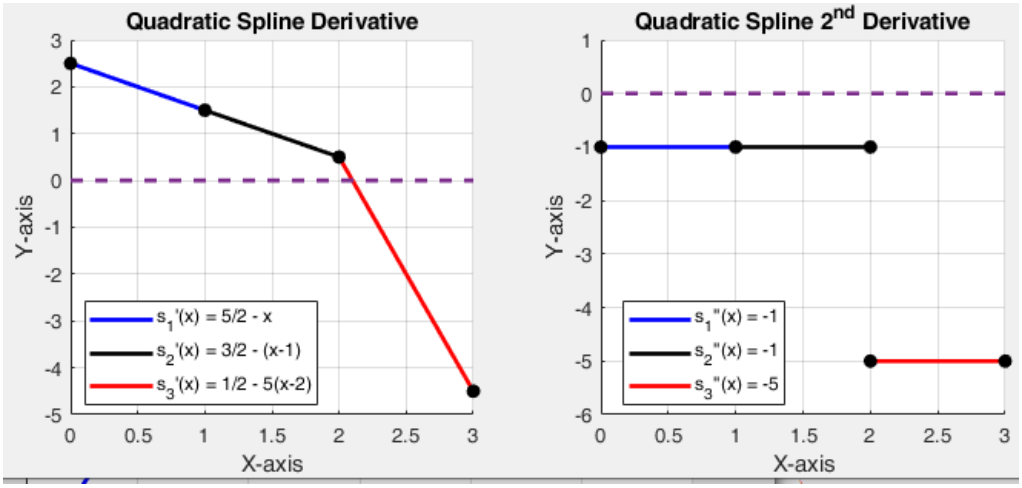
Let's combine all pieces together for our final answer, and plot:

$$\begin{cases} s_1(x) = \dfrac{-x^2 + 5x}{2}, & x \in [0, 1] \\ s_2(x) = \dfrac{4 - (x-1)^2 + 3(x-1)}{2}, & x \in [1, 2] \\ s_3(x) = \dfrac{6 - 5(x-2)^2 + (x-2)}{2}, & x \in [2, 3] \end{cases}$$



Finally, let's also plot the derivates, highlighting the general property of quadratic spline:

- The **1st derivative of a quadratic spline is continuous.**
- The **2nd derivative of a quadratic spline is discontinuous**.



Problem 3 (40pts) Numerical Differentiation

Consider the following 3-point finite difference approximation of the 1st derivative of smooth function $f(x)$:

$$f'(x)=\frac{-f(x+4h)+4f(x+2h)-3f(x)}{4h}+\text{Error}(h)$$

a) Use Taylor expansion to find the error in the expression above. Is this method first- or second-order accurate?

First, consider Taylor series:

$$\begin{cases} f(x+4h)=f(x)+4hf'(x)+\frac{(4h)^2}{2}f''(x)+\frac{(4h)^3}{3!}f'''(x) \\ \qquad\qquad =f(x)+4hf'(x)+8h^2f''(x)+\frac{32h^3}{3}f'''(x)+O(h^4) \\ f(x+2h)=f(x)+2hf'(x)+\frac{(2h)^2}{2}f''(x)+\frac{(2h)^3}{3!}f'''(\beta) \\ \qquad\qquad =f(x)+2hf'(x)+2h^2f''(x)+\frac{4h^3}{3}f'''(x)+O(h^4) \end{cases}$$

Multiply these by the coefficients in the given finite difference approximation, to simplify the final summation:

$$\begin{cases} -f(x+4h)=-f(x)-4hf'(x)-8h^2f''(x)-\frac{32h^3}{3}f'''(x)+O(h^4) \\ 4f(x+2h)=4f(x)+8hf'(x)+8h^2f''(x)+\frac{16h^3}{3}f'''(x)+O(h^4) \end{cases}$$

Note that the $f(x)$ terms and the quadratic terms will cancel when we plug these into the given expression:

$$\frac{-f(x+4h)+4f(x+2h)-3f(x)}{4h}=\frac{\left(-f(x)-4hf'(x)-\frac{32h^3}{3}f'''(x)\right)+\left(4f(x)+8hf'(x)+\frac{16h^3}{3}f'''(x)\right)-3f(x)+O(h^4)}{4h}$$

$$=f'(x)\underbrace{-\frac{4h^2}{3}f'''(x)+O(h^3)}_{\text{ERROR}}=f'(x)\underbrace{-\frac{4h^2}{3}f'''(c)}_{\text{If we used exact truncation}}:\ 2^{nd}\text{ order accurate}$$

b) Check your error expression by applying this finite difference approximation to the function $f(x)=x^3$.

$$\frac{-f(x+4h)+4f(x+2h)-3f(x)}{4h}=\frac{-(x+4h)^3+4(x+2h)^3-3x^3}{4h}$$

$$=\frac{\left(-x^3-12x^2h-48xh^2-64h^3\right)+4(x^3+6x^2h+12xh^2+8h^3)-3x^3}{4h}$$

$$=\frac{12x^2h-32h^3}{4h}=\boxed{3x^2-8h^2}$$

- Thus, we did obtain the derivative of this function, $(x^3)'=3x^2$.
- The error terms are also in agreement, since $(x^3)'''=6$, and therefore $-\frac{4h^2}{3}f'''(x)=-\frac{4h^2}{3}6=-8h^2$

Problem 3 (40pts) Differentiation

Use functions values $\boldsymbol{f(x-3h)}$, $\boldsymbol{f(x)}$, and $\boldsymbol{f(x+2h)}$ to find the finite difference for $\boldsymbol{f''(x)}$.

To do this, set up and solve the linear system of equations for constants (a, b, c) that results from expanding

$$af(x-3h)+b\,f(x)+c\,f(x+2h)$$

using a Taylor series around x up to 3rd order in *h*.

Don't forget to obtain the truncation error, as well. Is the expression you obtained first- or second-order accurate?

For both parts of the question, we will use Taylor expansion around x:

$$f''(x)\approx a\left[f(x)-f'(x)(3h)+f''(x)\frac{(3h)^2}{2}-f'''(x)\frac{(3h)^3}{3!}+\ldots\right]+bf(x)+c\left[f(x)+f'(x)2h+f''(x)\frac{(2h)^2}{2}+f'''(x)\frac{(2h)^3}{3!}+\ldots\right]$$

Collecting terms proportional to $f(x)$ and its derivatives at x, questions (a) and (b) require solving the following linear algebra problems:

$$\begin{cases} f(x): & a+b+c=0 \\ f'(x): & (-3a+2c)h=0 \\ f''(x): & (9a+4c)\dfrac{h^2}{2}=1 \\ f'''(x)(-27a+8c)\dfrac{h^3}{3!}=Error \end{cases}$$

$$\begin{cases} -3a+2c=0 \Rightarrow c=\dfrac{3}{2}a \\ 9a+4c=(9+6)a=\dfrac{2}{h^2} \Rightarrow \boxed{a=\dfrac{2}{15h^2}} \Rightarrow c=\dfrac{3}{2}a \Rightarrow \boxed{c=\dfrac{1}{5h^2}} \\ b=-a-c=-\dfrac{2}{15h^2}-\dfrac{1}{5h^2}=-\dfrac{5}{15h^2} \Rightarrow \boxed{b=-\dfrac{1}{3h^2}} \\ \text{Error}=(-27a+8c)f'''(x)\dfrac{h^3}{3!}=f'''(x)\left(-\dfrac{54}{15}+\dfrac{8}{5}\right)\dfrac{h}{3!}=\left(-\dfrac{18}{5}+\dfrac{8}{5}\right)f'''(x)\dfrac{h}{3!}=-2f'''(x)\dfrac{h}{3!}=\boxed{-\dfrac{f'''(x)\,h}{3}}+O(h^2) \end{cases}$$

Thus, our finite difference is

$$f''(x)\approx\frac{2f(x-3h)}{15h^2}-\frac{f(x)}{3h^2}+\frac{f(x+2h)}{5h^2}=\boxed{\frac{2f(x-3h)-5f(x)+3f(x+2h)}{15h^2}}\ \text{with error}=-\frac{f'''(c)\,h}{3},\ c\in(x-3h,\,x+2h)$$